Eesh Khanna

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LinkedIn | GitHub | Portfolio

Technical Skills

- **Programming, Data and Databases:** Python, SQL, Pandas, NumPy, Scikit-learn, PostgreSQL, MongoDB, Neo4j, FAISS, ChromaDB, Database Indexing and Caching
- **AI/ML, LLM and Quantum Computing:** PyTorch, TensorFlow, Hugging Face (FinBERT, SBERT), Llama 3, Claude APIs, RAG, GraphRAG, LangGraph, MCP, AutoML (TPOT), XAI (LIME), HMM, Monte Carlo Simulation, Time-Series (LSTM/GRU), PennyLane
- **Evaluation, MLOps and Backend:** DeepEval, Walk-forward Validation, Feature Engineering, AWS (EC2), Azure, Docker, Kubernetes, FastAPI, REST APIs, Load Testing, API Optimization, Git, GitHub

Work Experience

LumberFi Oct 2025 – Present
Software Developer Intern *Bengaluru, India*

- Researched QA bottlenecks across 50+ epics, identifying fragmented Jira context causing 3-hour manual test delays.
- Architected a 9-stage LangGraph multi-agent platform using Claude APIs, orchestrating persistent memory for test automation.
- Generated 500+ test cases, cutting authoring to less than 20m per epic, saving 100+ hours quarterly and boosting deployment 15%.
- Analyzed CSM onboarding delays, finding unstructured client data required 30-120 days of manual mapping to Lumber formats.
- Orchestrated agentic AI workflows to automatically parse, validate, and transform raw client data into Lumber-compatible schemas.
- Slashed onboarding from 30-120 days to 1-2 days, eliminating manual CSM data entry and accelerating client time-to-value.
- Researched Canadian payroll compliance, identifying gaps in multi-jurisdictional tax calculations and statutory reporting.
- Engineered a country-aware subsystem with deep-copy sync and optimistic locking, ensuring strict data integrity for tax logic.
- Enabled full CRA/T4 compliance across 13 provinces, processing 100% of statutory deductions and unblocking enterprise clients.
- Profiled backend bottlenecks via load testing, identifying N+1 queries and redundant fetches across 6 critical REST APIs.
- Optimized database indexing and caching, reducing average API latency from 4s to 500-600ms and increasing throughput 300%.

Dell Technologies May 2025 – Jul 2025
Agentic AI Intern *Bengaluru, India*

- Evaluated DCOS supply chain workflows and 13 week MongoDB shortage data, identifying critical manual demand reconciliation inefficiencies.
- Engineered a **6-agent ReAct platform** using Llama 3 and FastAPI to automate complex supply chain analysis.
- Integrated Model Context Protocol (MCP) to streamline data retrieval and automate end to end demand reconciliation.
- Reduced manual planning effort by **30%** and cut agent handling time by **15%** through intelligent automation.
- Ensured high system reliability and accuracy by rigorously validating agent performance using **DeepEval** regression testing frameworks.

Projects

Aviation Regulatory Intelligence Platform Live Demo | Github
Python, SBERT, FAISS, Llama, RAG, Streamlit

- Researched aviation regulatory compliance challenges, addressing the need for accurate, citation-backed insights for non-technical reviewers.
- Extracted actionable insights from over 4,000 unstructured aircraft modification records to address critical compliance gaps.
- Developed a robust **SBERT- and FAISS-based classification system** to ensure highly precise regulatory document categorization.
- Deployed a powerful **Llama-powered RAG pipeline** to automate complex regulatory intelligence retrieval tasks rapidly, seamlessly.
- Prepared an interactive Streamlit dashboard to streamline user access and significantly simplify overall system interaction.
- Achieved **0.82 Precision@5** and **0.91 Top-5 accuracy** for highly reliable, citation-backed regulatory intelligence data retrieval.
- Delivered reliable, citation-backed answers, significantly reducing manual document review time for non-technical compliance review teams.

Compliance Memory System: Agentic AI for Legal Compliance Live Demo | Github
Python, FastAPI, LangGraph, Neo4j, ChromaDB, DeepEval, Docker

- Analyzed legal compliance bottlenecks, identifying a severe lack of auditable, interconnected enterprise policy data sources.
- Highlighted critical inefficiencies found in manual regulatory gap detection processes within existing enterprise compliance systems.
- Architected a highly scalable **multi-agent GraphRAG system** utilizing **LangGraph, Neo4j, and ChromaDB** database technologies seamlessly.
- Implemented hybrid retrieval combining BM25, dense vector search, and advanced reranking for optimal search accuracy.
- Generated highly auditable, interconnected policy knowledge graphs directly from complex, unstructured legal and compliance documents.
- Enabled bidirectional compliance analysis and human-in-the-loop policy remediation workflows for all enterprise compliance end users.
- Automated regulatory gap detection to ensure robust, verifiable, and fully auditable enterprise-wide compliance business intelligence.

Publications and Patents

Quantum-Enhanced Forecasting Pipeline Integrating Sentiment Dynamics and Regime Modeling 2025
Expert Systems with Applications (ESWA) (In Proceedings, First Author) *Publication*

– Researched forecasting gaps; built a hybrid quantum-classical pipeline integrating FinBERT and HMM.
– Engineered regime-conditioned Monte Carlo simulations and a novel Quantum-Enhanced Adam optimizer.
– Reduced pooled RMSE by 38.9% across recurrent models, achieving an R2 of 0.9894 and a 4.43% MAPE.

Enhancing Explainability of AutoML and Deep Learning for Lumpy Skin Disease Detection 2025
Springer, ICAIC 2025 (Co-author) *Publication*

– Researched cattle disease detection; evaluated AutoML and deep learning models on epidemiological data.
– Applied Stratified K-Fold validation and SMOTE to handle class imbalance, ensuring robust generalization.
– Achieved 0.94 accuracy and 0.99 ROC AUC using TPOT, integrating LIME for transparent clinical insights.

Multi-Modal Disaster Verification from Geospatial and Text Data 2025
Patent Application No. 202511074034 A

– Researched disaster verification, identifying high false-positive rates in single-modality alerts.
– Built a multi-modal pipeline fusing SBERT text, KNN geospatial clustering, and real-time news.
– Achieved 85% accuracy and 0.92 specificity via adaptive scoring, minimizing false-positive alerts.
– Deployed containerized models with persistent caching, enabling near-instant report verification.

Hackathons

ZERO TO ONE AI Hackathon Finalist 2026
Pocket FM, OpenAI, Lightspeed *National Finalist*

– Researched script collaboration gaps; built a Git-like version control system with branches and PRs for version control.
– Engineered an AI agent to auto-resolve merge conflicts, cutting manual resolution time by 40%.
– Integrated Sora for visual scene previews and built a node-based narrative graph for structural analysis.
– Accelerated director feedback by 30% and improved narrative coherence via real-time visual mapping.

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Education

Manipal University Jaipur 2022 - 2026
B.Tech in Computer Science and Engineering *8.3/10 CGPA*